

Clinic Management System (Functions of Each Code)

1. Create Database: DROP DATABASE IF EXISTS clinic_management;

```
CREATE DATABASE clinic_management CHARACTER SET utf8mb4 COLLATE  
utf8mb4_unicode_ci;
```

```
USE clinic_management;
```

Function:

- Delete the old database if it already exists.
- Creates a new database called clinic_management.
- Sets UTF-8 encoding to support all characters.
- Select the database for use.

2. Department Table: CREATE TABLE Department (...)

Function:

- Stores all medical departments in the clinic system.
- Each department has: Unique ID and Department name
- dept_id is the Primary Key.

3. Clinic Table: CREATE TABLE Clinic (...)

Function:

- Stores clinic information.
- Includes: Clinic name, Address and Department ID
- Connected to the Department table using Foreign Key.
- One department can contain many clinics

4. Doctor Table: CREATE TABLE Doctor (...)

Function:

- Stores doctor data.
- Includes: Doctor name, Phone number, Address and Department
- Connected to Department table using Foreign Key.
- One department can have many doctors.

5. Patient Table: CREATE TABLE Patient (...)

Function:

- Stores patient information.
- Includes: Patient name, Phone, Address, Birth date and Job
- patient_id is the Primary Key.

6. Appointment Table: CREATE TABLE Appointment (...)

Function:

- Stores appointment details between patients and doctors.
- Includes: Appointment date, Start and end time, Cost, Status and Diagnosis
- Connected to: Patient table and Doctor table
- Uses Foreign Keys for relationships.
- Contains a CHECK constraint to ensure: end_time > start_time

Views

7. Appointment Details View: CREATE OR REPLACE VIEW vw_appointment_details AS

Function:

- Combines data from multiple tables into one virtual table.
- Displays: Appointment information, Patient name, Doctor name, Department and Clinic

- Simplifies data retrieval.

8. Patient Revenue View: CREATE OR REPLACE VIEW vw_patient_revenue AS

Function:

- Calculates: Total money paid by each patient and Number of appointments
- Uses: SUM(), COUNT() and GROUP BY

Insert Data

9. Insert into Department: INSERT INTO Department .

Function:

- Adds sample departments into the system.

10. Insert into Clinic: INSERT INTO Clinic

Function:

- Adds clinic records connected to departments

11. Insert into Doctor: INSERT INTO Doctor

Function:

- Adds doctor information into the database.

12. Insert into Patient: INSERT INTO Patient

Function:

- Adds patient records with personal information.

13. Insert into Appointment: INSERT INTO Appointment

Function:

- Adds appointment records between doctors and patients.
- Stores diagnosis, cost, status, and appointment timing.

Queries

14. Query 1: SELECT DISTINCT p.pat_name ...

Function:

- Displays patients diagnosed with fatty liver during the last year.

15. Query 2: SELECT c.clinic_name, c.address ...

Function:

- Displays addresses of cardiology clinics.

16. Query 3: SELECT p.pat_name, SUM(a.cost) ...

Function:

- Calculates total amount paid by patient ID 12527 in the last three years.

17. Query 4: SELECT d.dept_name, COUNT(a.appt_id) ...

Function:

- Counts number of appointments in each department.

18. Query 5: SELECT doc.doctor_name ...

Function: Displays: Doctor name, Department and Number of treated patients

19. Query 6: SELECT a.appt_id ...

Function:

- Displays all postponed appointments with patient and doctor information.

20. Query 7: SELECT diagnosis, COUNT(*) ...

Function:

- Displays the most common diagnoses in the system

